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RECORDING

Guide to Home Recording

Everything you need to know to get professional results from your home studio setup.

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Quality recordings begin long before mixing.

> Even with the best mixing or mastering engineer, a track can only sound as good as the recordings allow. Capturing clean, balanced, and expressive performances at home is entirely possible — it just requires attention to detail and a few smart choices. This guide will help you get professional-sounding results from your home setup.

1. Room and Environment

Choose the Right Space

Pick the quietest room possible — one without noticeable echo or outside noise. Avoid rooms with large reflective surfaces (like bare walls and windows). Soft furnishings, rugs, and curtains help reduce reflections and make the space more controlled.

If possible, avoid recording near computers, air conditioners, or other noise sources.

> Tip: Clap in the room — if you hear a harsh, metallic echo, it’s too reflective. Add some blankets or acoustic panels.

Acoustic Treatment

You don’t need a professional studio build. A few simple measures go a long way:

- Hang thick curtains or duvets behind and to the sides of your microphone.
- Place a rug underfoot if the floor is hard.
- Use a reflection filter behind the mic if available.

The goal is to minimize sound bouncing around the room before it hits the microphone.

2. Microphone Technique

Positioning and Distance

Keep about **15–20 cm (6–8 inches)** between your mouth and the microphone.
Use a **pop filter** to reduce plosive sounds (like “p” and “b”).
Stay consistent in your position — don’t move forward or backward mid-take.

If you’re using a condenser mic, aim it slightly off-center to reduce sibilance (harsh “s” sounds).

Gain Staging

Set your recording levels so that peaks hit around **-12 dBFS to -6 dBFS** on your input meter.
Avoid clipping (red lights or flat-topped waveforms). Digital clipping cannot be repaired later.

> If your takes are too quiet, raise the input gain slightly — but never at the expense of distortion.

Monitoring

Use **closed-back headphones** when recording vocals or live instruments. This prevents the backing track from bleeding into your microphone.
Keep the headphone volume moderate to avoid feedback or performance fatigue.

3. Recording Instruments

Drums

> A drum kit's sound is mostly decided before a single mic goes up. Sizes, heads, and tuning shape the tone far more than mic choice or placement ever will.
> Once the kit sounds right in the room, mic'ing it well is mostly about capturing each piece for what it's actually there to do, not trying to make every mic sound like the whole kit.
> Drums are also the one instrument where editing has to happen first, since everything else in the song will end up locked to their timing.

Start With the Kit, Not the Mics

Size and head choice affect the tone more than most people expect:

- **Snare depth changes the bottom end.** A deeper shell gives a fuller, louder snare; going shallower trades some of that low end for a tighter, quicker sound. Shell material matters too: metal shells tend to add high-end bite and cut, wood shells run a bit warmer.
- **Bigger cymbals resonate more and bleed further into other mics.** Live, that projection is useful. In a treated room, smaller, thinner cymbals are easier to control and cause less trouble later, though it's ultimately down to feel and preference.
- **Kick depth affects punch versus boom.** A very deep, wide kick shell resonates more than a small room can handle cleanly. For a tight, modern, punchy low end, a shallower kick, commonly around 45–50 cm (18–20 inches) deep, is usually easier to work with than a huge one.

If a drummer already has their kit dialed in and sounding good, there's rarely a reason to change much. Step in with tuning or head swaps mainly when it's needed, not as a default.

Replace Heads and Tune Before You Record

Fresh heads read brighter and behave more predictably than heads that are already stretched out or unevenly tuned, so it's worth doing this right before a session rather than days ahead. The one common exception is the kick's batter head, which usually holds its tone and tuning for a long time and doesn't need replacing every session.

A few general tuning principles:

- **Toms:** tune the bottom (resonant) head before putting the top head back on, tapping around all the lugs and listening for an even pitch all the way around. For a punchy, controlled tone with less ring, tune the bottom head relatively higher than the top. For a more open, resonant tone, tune the two heads closer together.
- **Snare:** both heads generally want to be tight, for crack on top and thud underneath, again checking that the pitch is even around all the lugs. Don't chase all the ring out of it, though — a little natural sustain gives you more character to work with once it's sitting in a full mix later.
- **Kick:** loosen both heads until they're barely holding tension, then bring them up just slightly. Combined with some internal muffling (a blanket or pillow resting against the heads), that gives a tight, slappy low end with minimal ringing.

Tuning by ear is mostly trial and error: try it a bit tighter, listen, and back off if it sounds worse. If a head keeps ringing more than you want after it's tuned, taming it with a small piece of tape or a dampening gel near the edge is an easy fix, but avoid deadening it completely. Some natural ring is easier to shape later than a sound that's already been squeezed lifeless.

Listen in the Room Before You Set Up a Single Mic

Once the kit is tuned, get the drummer to play through a section while you walk around the room and just listen. Fix anything that stands out, too boomy, too ringy, uneven between drums, before you commit to mic placement. A mic captures whatever's already happening in the room; it's not a tool for fixing a kit that doesn't sound right yet.

Mic Each Piece for What It's There to Capture

A useful mindset for placement: most mics on a kit are there to capture one specific piece cleanly, not the whole kit at once.

- **Kick:** placed near the beater's strike point, slightly off-center rather than dead-on. Pulling the mic back brings in more low end and room tone; moving it closer to the beater brings in more attack and less boom.
- **Snare (top and bottom):** most mics used here are directional, picking up mostly from the front and rejecting sound from behind. Use that to your advantage by angling the mic so hi-hats and cymbals fall into its rejection zone as much as possible. Aim the top mic at the strike point, positioned 2–3 cm (about an inch) above the rim. Mirror that on the bottom, angled toward the center but sitting a little farther back so it isn't overwhelmed by the snare wires.
- **Toms:** similarly, 2–3 cm (about an inch) above the rim, aimed at the strike point, and tucked between cymbals where possible so they help shield the mic from spill.

- **Overheads and cymbal mics:** the goal here is usually to capture the cymbals clearly, not to get a whole-kit picture, since close mics and a room mic are already handling the rest. Mic them fairly close, and aim slightly toward the edge of the cymbal rather than dead center for a smoother, less harsh tone. If you're running a stereo pair, measure from the center of the snare to each mic and keep that distance equal; that keeps the snare centered and phase-coherent between the two, instead of drifting to one side.
- **Hi-hat:** angle the mic off-axis a bit rather than pointing it straight on. A mic aimed directly into hard-played hi-hats can pick up the actual rush of air off the cymbals, which reads as an odd, unwanted sound. Keep it clear of where the stick travels, too.
- **Room mic(s):** how far back to place these depends entirely on your space. A large, lively room can support a distant pair for real ambience; a small or dead-sounding home room usually does better with a mic pulled in much closer, or just a single room mic rather than a stereo pair.
- **A "character" mic, if you have a spare input, is worth trying.** Stick it somewhere unconventional, a hallway, behind furniture, near a doorway, anywhere with an interesting sound. It won't always make the final mix, but occasionally it becomes the thing that defines the whole drum sound.

Label your tracks clearly inside your DAW:

- `Kick In`
- `Snare Top`
- `Snare Bottom`
- `Rack Tom`
- `Floor Tom`
- `Hi-Hat`
- `Overhead L`
- `Overhead R`
- `Room`

Check Phase Before You Track

Once mics are up, sum everything to mono and compare each close mic against the overheads one at a time, flipping phase on each and keeping whichever version sounds fuller, with more low end. The bottom snare mic is the one that almost always needs flipping, since it's facing the opposite direction from the top mic relative to the head's movement. Don't get stuck on this step; phase can still be flipped later if something sounds off once you're mixing.

Keep the Signal Chain Simple While Tracking

For most tracks, a clean preamp with healthy gain staging and no processing is enough. If you want to shape a sound going in, light compression on the snare, a moderate ratio with a slower attack that lets the initial hit through, can add snap and keep levels consistent, but keep it subtle so you're not locked into a decision you can't undo later. A room or character mic is the one place it's fine to be heavier-handed, since pushing it harder (more sustain, more compression) can add grit that's difficult to recreate afterward.

Capture Safety Samples Before Full Takes

Before tracking full performances, get the drummer to play a few isolated hits of each piece, kick alone, snare alone, each tom, each cymbal, while the kit is freshly tuned. These rarely get used, but they're there if a head starts sounding inconsistent later in the session, or if you need a clean hit to reinforce a weak moment while editing or mixing.

Track in Big Sections, Then Comp and Punch In

Aim for full takes, or the largest sections the drummer can play through cleanly, and get more than one so you can comp the strongest pieces into a final performance rather than relying on a single pass. Listen for energy as much as timing: a drummer who's steady with the click but tiring and hitting softer over a long take is a bigger problem than a minor timing slip you can fix later.

When comping between takes, cut on a clear hit, kick, snare, or crash, rather than in the middle of a phrase; it's far more forgiving to splice cleanly. For any punch-ins, always roll in a few bars early so the drummer is locked into the feel before the punch point, and pay closest attention to how the cymbals match on either side of the edit. An inconsistent cymbal choice or dynamic right at a punch is usually the easiest way to give away an edit.

Edit the Drums Before You Track Anything Else

Whatever gets tracked next, bass, guitars, keys, will end up locked to the drums, so finish editing them first. Editing drums after other instruments are already recorded against them is far more painful than doing it up front.

- **Grid-edit off the right tracks.** Quantize based on the kick, snare, toms, and hi-hat, but leave the overhead and room tracks to just follow along rather than triggering edits directly off them; cymbal transients are much harder for any tool to detect accurately.
- **Favor the snare over the kick when they land together.** When a kick and snare hit at the same time but aren't quite aligned, nudge the kick and leave the snare closest to the grid. It tends to read as tighter, since a kick that's slightly off on its own is far less noticeable than a snare that is.
- **Handle odd groupings by hand.** Triplet fills and other non-standard groupings usually trip up automatic quantizing tools. Switch to the matching grid and place those hits manually rather than fighting the tool to get it right.
- **Crossfade every edit point**, and if a fade is clipping a cymbal decay or causing a flammed, doubled-sounding hit, nudge the cut to a different nearby hit instead of forcing it.
- **Clean up once the edits are solid.** Consolidate each track down to a single region per part, and clear out the unused audio regions left behind from editing so the session stays light and easy to work in.

Rough-Mix the Drums Before You Move On

Once drum tracking wraps, spend a few minutes getting them close to a finished-sounding tone, some EQ, light compression, maybe a sample layered under the snare, before moving on to the next instrument. It's not the final mix, but a drum sound that already feels close to finished keeps the session inspiring and gives you a far more realistic target for how everything else should sit against it.

What to Deliver to Your Mixer

A finished drum session handed off for mixing should include:

- Comped, edited performances rather than a single raw pass with mistakes left in
- One consolidated, labeled track per mic or piece, not dozens of leftover regions and crossfades from editing
- Isolated safety samples for the kick, snare, and cymbals
- Consistent levels with no clipping, and phase already checked across the close mics and overheads
- A rough mix reference, so the mixer understands the direction you were going for

> If the kit isn't tuned well or the takes aren't tight, no amount of mixing fixes that afterward. Solving these problems is cheap while you're tracking. Fixing them later is expensive, if it's even possible at all.

Electric Guitar

> Always check your tuning before every take. Guitars drift faster than you think.
> Even minor pitch shifts between double-tracked guitars can ruin tightness and stereo image.
> Fresh strings and frequent tuning make the mix and performance sound instantly more professional.

Get Your Guitar Set Up Right

Most tuning problems start before anyone hits record. Get these three things right first:

- **Match string gauge to your tuning.** Strings too light for a low or dropped tuning lose stability under hard picking, sagging in pitch right after the attack. Go a gauge or two heavier: around 11–56 for half-step-down or drop D, 12–60 for drop C or lower, and a heavy single (upper 60s to low 70s) or baritone set for drop A and below.
- **Restring right before the session, not days ahead.** An older, stretched string drops pitch noticeably as a note rings out, and it only gets worse with age. Bring new strings up to pitch without aggressively stretching them by hand; you want stable tuning for a few hours, not a “broken-in” set for a live show.

- **Use a tuner that reads in cents, and check intonation too.** A cheap clip-on tuner's in-tune range is often wider than it looks. For intonation, play the same note open, then at the 12th fret. If the fretted note is sharp or flat against the open string, that's a setup issue (bridge saddle adjustment), not a tuning one, and it should be checked at the tuning you're actually recording in.

Play Tight, Then Edit Tight

No amount of mixing rescues a sloppy performance. A few habits make this much easier:

- **Rehearse until it's genuinely comfortable, before you record.** Inconsistent picking dynamics, buzzing frets, and timing slips are far easier to fix by playing the part again than by editing around them afterward.
- **Tune at the same intensity you're actually going to play with.** A string tuned while picking softly reads sharp the moment you dig in for real, especially on a low string. Picking closer to the bridge also helps. Retune often too, even if the last take sounded fine, since a guitar can drift as a whole and still sound in tune with itself.
- **Punch in on just the bad bar**, not a whole shaky pass because "the rest was good." For a genuinely difficult passage, it's fine to isolate it: loop just that phrase against a click until it's clean, then drop it in. Treat that as a last resort though, not your default. A slightly imperfect take with real character almost always beats one that's technically flawless but robotic.

Record in Sections, Not One Take

Trying to nail an entire song in a single unbroken pass almost always produces uneven energy and creeping tempo or pitch drift, especially past the two-minute mark.

- **Record in pieces.** Track verse, chorus, and bridge separately, or phrase by phrase for tricky riffs, then comp together the strongest takes. This is standard practice on professional sessions, and it's the difference between a track that sounds *played* and one that sounds *performed*.
- **Don't pile on layers for size.** Two genuinely tight, clean rhythm takes (left and right) usually sound bigger than four or five loose ones, since it's the tightness between takes that creates size, not the track count. If a part repeats identically later in the song, copy the take you already have instead of re-recording it.

Use a Real Amp or a Proper Amp Simulator

If you're tracking direct through an interface, multi-effects unit, or modeler, confirm that an actual amp simulator is engaged before you print the signal: a modeled amp and cabinet, not just compression, EQ, and time-based effects. Guitar run straight into an interface without amp/cab modeling has no speaker breakup or power-amp saturation shaping it. It reads as thin, harsh, and unmistakably "digital," no matter how good the playing is.

> Before committing to a full session, record a short test pass and listen back on headphones. If it doesn't sound like a mic'd amp, your amp sim isn't actually in the signal path. Check your patch or plugin chain before you track anything for real.

Always Print a DI Track

Whether you're mic'ing a real amp or recording direct, always capture a clean **DI signal** in parallel with the processed tone.

- **It's a safety net.** This costs nothing during tracking but gives your mixer the option to re-amp or change the tone entirely without needing you back in the room.
- **It makes editing far easier.** A processed guitar track is often just a solid block of waveform, but the DI shows exactly where each note and strum starts, which matters if timing ever needs tightening. A processed-only take with no DI is a dead end if the sound needs to change later.

Label your tracks clearly inside your DAW:

- `Guitar Rhythm DI`
- `Guitar Rhythm Amp`

Give Leads Their Own Identity

A lead using the exact same guitar, tone, and settings as the rhythm tracks tends to blend in rather than cut through.

- **Change something about the tone.** Swapping guitars, or even just pickups, through the same amp setup is often enough to separate a lead from the rhythms underneath it. A touch more midrange helps it sit on top of a dense mix too.
- **Mute unused strings.** For single-note lines and octaves, lightly muting strings you're not playing with your fretting or picking hand cuts down on noise that gets buried under rhythm guitars but sticks out on an exposed lead.

Clean Up Your Edits Before You Deliver

Before a session is really done, whether you're mixing it yourself or sending it off, go back through every guitar track in solo.

- **Check every edit and crossfade** for clicks, pops, or a fade that got skipped while you were moving fast during tracking.
- **Don't snap everything to the grid.** Nudge anything that's a touch early or late, but over-tightened edits lose their feel and start to sound robotic. A little natural variation is what keeps it sounding human.
- **Trim ringing tails** so a new section hits cleanly instead of bleeding into what came before it, and keep a duplicate of your edited tracks before consolidating them down to one clean region per part.

What to Deliver to Your Mixer

A finished guitar session handed off for mixing should include:

- In-tune, edited performances comped from multiple takes, not a single raw pass with mistakes left in

- A clean DI track for every part, printed alongside any amp or amp-sim tone
- At least one reference tone print (mic'd amp or amp sim) so the mixer understands the direction you were going for
- One consolidated, edit-checked track per part, not dozens of leftover regions and crossfades from tracking
- Consistent levels with no clipping, and noise or silence trimmed between sections

> If a part isn't in tune, isn't tight, or was recorded with no amp sim and no DI, no mix can fully fix it. Solving these problems is cheap during tracking. Fixing them afterward is expensive, if it's even possible at all.

Bass

> Bass tuning is simpler than guitar: one string, one note at a time. But that also means there's nowhere to hide a bad note.
> A few extra minutes on setup and signal chain save hours of trying to fix a bass part that's out of tune or lost in the mix.
> Unlike guitar, bass doesn't need to be edited perfectly tight. A little natural feel against a locked-in grid is usually a good thing.

Track Guitars First

Tuning is the hardest thing to keep airtight on guitar. Technique, strings, and setup all fight against you, while bass is comparatively simple, since it's one note at a time.

Recording all the guitars first means that if a chord somewhere on the neck ends up a touch sharp or flat, you can nudge the bass note to match it during bass tracking, a small, easy fix. Do it the other way around and a slightly-off bass makes it nearly impossible to match a full chord shape to later. It's also easier to shape a bass tone that sits well under the guitars once they already exist, rather than guessing at how it'll fit beforehand.

Get Your Bass Set Up Right

The same setup principles as guitar apply, just scaled up.

- **Go heavier on string gauge for lower tunings.** A half-step or so down, a heaviest-available four-string set (roughly a **.110** low string) is a reasonable baseline. Anything lower, go to a five-string with heavier gauges still.
- **Check intonation the same way as guitar** (open string against the 12th fret). If it reads sharp with no more room to move the saddle back, try lowering the string action instead; less finger pressure needed to fret a note means less pitch-sharpening from the press, and it often solves what the saddle alone can't.

Bass Can Sit a Little Looser

Once the guitars are locked down tight and the drums are gridded, a bass played a touch more naturally actually complements the rest of the track rather than fighting it. Bigger sections and less note-by-note editing are fine here. Aim for solid takes and good tuning, but bass doesn't need the same tightness as rhythm guitar.

A few things worth watching for:

- **String ringing.** The low string can bleed into the next note as the player jumps between strings, a common bass habit that reads as messy against a tight mix. Mute the previous string, or split the take into smaller punches if it keeps happening.
- **High notes on a low string.** For a note fretted high up on a lower string, it's often easier to get accurate tuning by punching that note in on the next string up instead.
- **Fret-specific tuning issues.** Problems tend to show up at particular spots, like the first fret of a string. If a part lives there, check the tuning right at that fret rather than trusting a general tune-up.

Blend a Clean and a Dirty Tone

You don't need to mic a bass cab to get a great bass tone. A clean DI plus a driven or saturated tone (from a bass preamp/drive pedal or plugin) is a complete, professional-grade approach on its own.

If you're layering more than one source, each one has a job:

- **DI:** stays clean and flexible, and always gives you the option to re-amp or change the tone later without calling anyone back in.
- **Driven layer:** adds midrange aggression that helps the bass cut through dense guitars.
- **Mic'd amp** (if you have one): sits underneath as the heaviest, dirtiest of the three.

Two things to watch when blending them: check that a clean DI and a processed layer are in phase, since having traveled through different paths they can land slightly out of alignment and cancel some frequencies (listen for whichever phase relationship sounds fuller). And don't expect the dirty layer to carry the low end; it's there for midrange bite, so it's fine, often better, to roll off some of its low and extreme high end.

Label your tracks clearly inside your DAW:

- `Bass DI`
- `Bass Drive`
- `Bass Amp`

Clean Up Your Edits Before You Deliver

Before a session is really done, whether you're mixing it yourself or sending it off, go back through the bass track in solo and check each edit and crossfade for clicks, pops, or a missed fade.

- **Judge the edits in context**, against the rest of the rhythm section rather than in solo, so you don't over-tighten something that already sits well with the drums.

- **Reuse clean repeats.** If a section repeats and an earlier pass is noisier or looser than a later one, it's fine to copy the cleaner repeat into place rather than fixing the messier one by hand.
- **Keep a duplicate** of your edited tracks before consolidating them down to one clean region, so you can always get back to the individual takes later.

What to Deliver to Your Mixer

A finished bass session handed off for mixing should include:

- In-tune, edited performances comped from multiple takes, not a single raw pass with mistakes left in
- A clean DI track, printed alongside any driven or amp tone
- At least one reference tone print (driven or mic'd amp tone) so the mixer understands the direction you were going for
- One consolidated, edit-checked track, not dozens of leftover regions and crossfades from tracking
- Consistent levels with no clipping, and noise or silence trimmed between sections

> If a part isn't in tune, isn't tight, or has no DI backing up the tone you printed, no mix can fully fix it. Solving these problems is cheap during tracking. Fixing them afterward is expensive, if it's even possible at all.

Acoustic Instruments

Use a condenser mic about 30–60 cm away, aimed where the neck meets the body for a balanced tone.
Avoid pointing the mic directly at the sound hole — this can cause boomy low-end buildup.

Record multiple takes and choose the most consistent performance, not necessarily the first.

MIDI and Virtual Instruments

If you're using software instruments, always **export the MIDI performance and a rendered audio version** once your production is complete.
This ensures the mix engineer can reproduce or tweak sounds later if needed.

Inside your DAW, keep naming consistent:

- `Piano Main`
- `Synth Arp`
- `Pad Texture`

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4. Vocal Recording

Preparation

Warm up your voice, hydrate, and rehearse before recording.
Avoid dairy, carbonated drinks, or coffee just before vocal takes — they can affect tone and clarity.

Performance

Focus on **emotion and consistency** over perfection.
If you're recording harmonies or doubles, keep phrasing and timing tight — but let natural differences add depth.

Record multiple takes (3–5) for each section. Label them clearly in your DAW:

- `Lead Vocal Take 1`
- `Lead Vocal Take 2`
- `Lead Vocal Comped` (after editing)

Comping and Editing

Select the best phrases from each take and compile them into a single, strong performance.
Remove clicks, coughs, or long silences.
Avoid excessive tuning or timing correction — keep it natural unless the style demands precision.

5. File Management and Organization

Folder Structure

Keep your project tidy from the start.
A clear folder system avoids confusion when you later export for mixing.

Example:

```
/SongTitle_Session/
├── Audio/
│   ├── Lead Vocal Takes/
│   ├── Guitars/
│   ├── Bass/
│   ├── Drums/
│   └── Synths/
├── MIDI/
│   ├── Piano Main.mid
│   └── Synth Arp.mid
├── RoughMixes/
│   └── SongTitle_RoughMix.wav
└── ProjectFiles/
    └── SongTitle.logicx
```